Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 42 Electrical Power

1) A light bulb operating at a dc voltage of 120 V has a resistance of 200 Ω. How much power is dissipated in this bulb?

$P = \frac{V^2}{R} = 72\,W$

2) A 150 W car light runs on a 12 V battery, what is the resistance of the car light?

$P = \frac{V^2}{R} \quad | \quad R = \frac{V^2}{P} = 0.96\,\Omega$

3) A 100 W heater is connected to a 110 V source. What current flows through the heater?

$P = IV \quad | \quad I = \frac{P}{V} = 0.91\,A$

4) A 100 W motor is run using a 12 V power source,

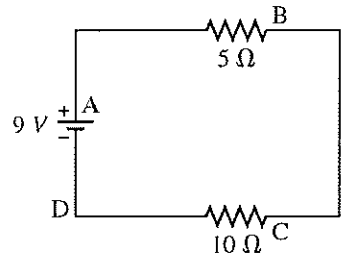
A) What resistance does the motor have?

$P = \frac{V^2}{R} \quad | \quad R = \frac{V^2}{P} = 1.44\,\Omega$

B) What current does the motor draw from the power supply?

$P = IV \quad | \quad I = \frac{P}{V} = 8.3\,A$

5) A 9-V battery is hooked up to two resistors in series. One has a resistance of 5 Ω, and the other has a resistance of 10 Ω. Several locations along the circuit are marked with letters, as shown in the figure. Through which resistor is energy being dissipated at the higher rate?



(A) the 10-Ω resistor
B) the 5-Ω resistor
C) Energy is being dissipated by both resistors at the same rate.

$P = I^2 R$ Same Current

Larger $R \rightarrow$ Larger P

6) A 100-W lamp glows brighter than a 25-W lamp. The electrical resistance of the 100-W lamp is

(A) less.
B) greater.
C) the same.

$P = \frac{V^2}{R}$ | Same V

$R = \frac{V^2}{P}$ | Larger $P \rightarrow$ smaller R

7) What is the peak voltage in an ac circuit where the rms voltage is 120 V?

$V_{rms} = \frac{V_p}{\sqrt{2}}$ | $V_p = \sqrt{2}\, V_{rms} = 169.7\,V$

8) An ac power source runs at a frequency of 50 Hz and have a peak volage of 50 V, if a 100 Ω motor is connected to this power source,

A) What is the rms current through the motor?

$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{50V}{\sqrt{2}} = 35.35V$ | $I_{rms} = \frac{V_{rms}}{R} = \frac{35.35V}{100\Omega} = 0.35A$

B) What is the average power used by the motor?

$P_{avg} = I_{rms} V_{rms} = (0.35A)(35.35V) = 12.5\,W$

9) A 150 W lamp is plugged into an outlet that runs at a peak voltage of 170 V. What is the rms current through the lamp?

$V_{rms} = \frac{V_p}{\sqrt{2}} = 120.2V$ | $P = I_{rms} V_{rms}$ | $I = \frac{P}{V} = 1.25A$

10) In order to has the exact same output power for a light bulb using both ac or dc current, what would the peak ac current be if the dc current was 5 A?

$I_{rms} = I = 5A$ | $I_{rms} = \frac{I_p}{\sqrt{2}}$ | $I_p = \sqrt{2}\, I_{rms} = \sqrt{2}(5A)$

$I_p = 7.07A$

11) The ac current through a 50 Ω resistor is $I = (0.80\ A)\sin(240t)$, where t is measured in seconds. What is the rms current through the resistor?

$I = I_p \sin(2\pi f t)$

$I_p = 0.8A$

$I_{rms} = \frac{I_p}{\sqrt{2}} = 0.57A$

12) A blender is connected across an ac voltage source given by the equation below, if the rms current that the blender draws is $I_{rms} = 2$ A, what is the power required by the blender?

$P = I_{rms} V_{rms}$

$V = (300\ V)\sin[2\pi(60\ Hz)t]$. $\quad 377/s$

$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{300V}{\sqrt{2}} = 212V$ | $P = (2A)(212V)$

$P = 424\,W$

13) From the above problem, what is the frequency of the ac current through the 50 Ω resistor?

$2\pi f = 377/s$

$f = \frac{377/s}{2\pi} = 60\,Hz$

14) The most efficient method for transmission of electrical energy across power lines is

A) High current
B) High voltage
C) Low voltage
D) Both A and B
E) None of the above

15) Which electrical property of a power source is most detrimental to the human body?

A) Resistance
B) Current
C) Voltage
D) None of the above

KEY

1) 72 W
2) 0.96 Ω
3) 0.91 A
4) A) 1.44 Ω B) 8.3 A
5) A
6) A
7) 169.7 V
8) A) 0.35 A B) 12.5 W
9) 1.25 A
10) 7.07 A
11) 0.57 A
12) 424 W
13) 60 Hz
14) B
15) B